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SKILL : WEEK 6

Task -1: To Store Commands, Maintain History Buffer, Handle Capacity Limits, Retrieve Entries, Display History, Validate Consistency

```
LM PICO 5.09 File: hj.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX_HISTORY 5
#define MAX_COMMAND 100

char history[MAX_HISTORY][MAX_COMMAND];
int history_count = 0;

/* Add command to history */
void add_command(const char *command)
{
    int i;

    /* If buffer is full, remove the oldest command */
    if (history_count == MAX_HISTORY)
    {
        for (i = 1; i < MAX_HISTORY; i++)
        {
            strcpy(history[i - 1], history[i]);
        }
        history_count--;
    }

    strcpy(history[history_count], command);
    history_count++;

    printf("Command stored successfully.\n");
}

/* Display history */
void display_history()
{
    int i;

    if (history_count == 0)
    {
        printf("History is empty.\n");
        return;
    }

    printf("\nCommand History:\n");

    for (i = 0; i < history_count; i++)
    {
        printf("%d %s\n", i + 1, history[i]);
    }
}

/* Retrieve a particular history entry */
void retrieve_command(int number)
{
    if (number < 1 || number > history_count)
    {
        printf("Invalid history entry.\n");
        return;
    }
}

Get Help      WriteOut      Read File      Prev Pg      Cut Text      Cur Pos
Exit          Justify        Where is       Next Pg      UnCut Text    To Spell
```

Output:

```
nrk@nallas-mac-Book-Air ~ % nano hj.c
nrk@nallas-mac-Book-Air ~ % gcc hj.c -o hj
nrk@nallas-mac-Book-Air ~ % ./hj

===== COMMAND HISTORY =====
1. Store Command
2. Display History
3. Retrieve Entry
4. Validate History
5. Exit
Enter your choice: 1
Enter command: 1
Command stored successfully.

===== COMMAND HISTORY =====
1. Store Command
2. Display History
3. Retrieve Entry
4. Validate History
5. Exit
Enter your choice: 5
Exiting program...
nrk@nallas-mac-Book-Air ~ %
```

Task -2: To Create Pipeline Structures, Store Commands, Maintain Execution Order, Connect Processes, Validate Pipeline Layout, Test Configurations
Code:

```
LM PICO 5.09 File: bb.c

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_COMMANDS 5
#define MAX_ARGS 20
#define MAX_LENGTH 100

/* Split a command into arguments */
void parse_command(char *command, char *args[])
{
    int i = 0;
    char *token;

    token = strtok(command, " \\t\\n");

    while (token != NULL && i < MAX_ARGS - 1)
    {
        args[i++] = token;
        token = strtok(NULL, " \\t\\n");
    }

    args[i] = NULL;
}

/* Validate pipeline */
int validate_pipeline(char *pipeline)
{
    int length = strlen(pipeline);

    if (length == 0)
    {
        printf("Error: Empty pipeline.\n");
        return 0;
    }

    if (pipeline[0] == '|' || pipeline[length - 1] == '|')
    {
        printf("Error: Pipeline cannot start or end with '|'.\n");
        return 0;
    }

    if (strchr(pipeline, "|") != NULL)
    {
        printf("Error: Empty command found in pipeline.\n");
        return 0;
    }

    return 1;
}

int main()
{
    char pipeline[MAX_LENGTH];
    char *commands[MAX_COMMANDS];
    int command_count = 0;
    int i;

    Get Help
    Exit
    WriteOut
    Justify
    Read File
    Where is
    Prev Pg
    Next Pg
    Cut Text
    UnCut Text
    Cur Pos
    To Spell
```

Output:

```
mrk@nallas-mac-Book-Air ~ % nano bb.c
mrk@nallas-mac-Book-Air ~ % gcc bb.c -o bb
mrk@nallas-mac-Book-Air ~ % ./bb
Enter pipeline command:
Example: ls -l | grep .c | wc -l
> █
```